Map memorization:

Bring up a single robot for mapping: ros2 launch hawkbot bringup_launch.py ip:=192.168.88.205 (Change according to the robot's IP address)

Start mapping with Rviz2: ros2 launch hawkbot_cartographer cartographer.launch.py If drift occurs, start over.

Save the map in .pbstream format: ros2 service call /write_state cartographer_ros_msgs/srv/WriteState "{filename: '/home/hawkbot/ROS2_WS/src/hawkbot/map/hawkbot.pbstream', include_unfinished_submaps: true}" (Change according to the file path)

Starting navigation:

Bring up multiple robots: ros2 launch hawkbot multi_bringup_launch.py ip1(change according to the desired robot number) :=192.168.88.205 ip2:=192.168.88.183

Launch Rviz: ros2 launch hawkbot_navigation2 multi_navigation_launch.py

Set local Costmap to /static_map




Set Trainee Bleue (robotx) to /robotx/recorded_trail_marker



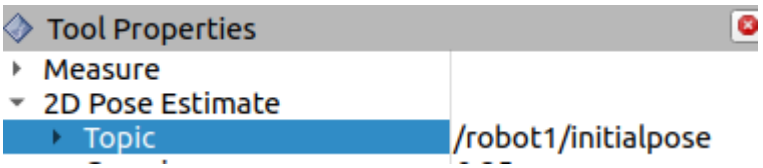

Set Cible Parcours (robotx) to /robotx/target_waypoint_marker



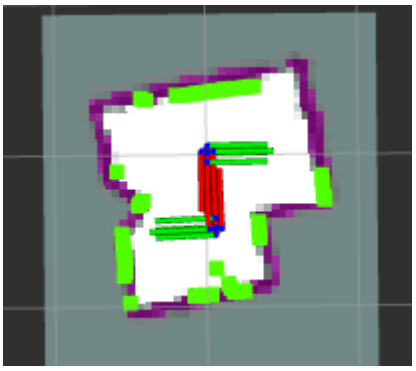

For initialpose calibration, switch between the robotx/scan topics in LaserScan



For initialpose, go to Panels > Tool Properties > 2DPoseEstimate and modify the topic to choose the robot whose position you want to initialize




Try to position your robot as close as possible to its real-world location by making sure to overlay the static map and the green points from the scanner



Navigation program:

drive robotx with: ros2 run hawkbot teleop_keyboard robotx

Draw a shape on the ground then ask robotx to follow the path : ros2 run hawkbot trajectory_control_menu

Monitor manually Robot1 and ask Robot2 to follow Robot1 : ros run hawkbot follow_leader_menu